

Denoise First, Orthogonalize Later

Understanding Momentum in Muon via Spectral Filtering

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TL;DR

Muon shows strong empirical performance in LLM training, but the theoretical role of momentum remains unclear. We show that momentum in Muon acts as a **spectral filter**: under a **signal-plus-perturbation** gradient model, it suppresses perturbations while preserving the dominant gradient signal, **enlarging the spectral gap** between them and stabilizing the singular subspaces passed to Muon's orthogonalization step. Applying momentum **before** orthogonalization achieves provably stronger signal alignment than reversing the order or removing momentum — hence the rule: **denoise first, then orthogonalize**.

- **Spectral gap** grows with the momentum window size $T = \frac{1}{1-\beta}$; the top- r subspace error decays as $O(T^{-1/4})$.
- **Validated** on synthetic, CIFAR-10, and NanoGPT probe experiments.

1 The question Muon raises

Muon and Orthogonal-SGDM use the **same two operations** — momentum and orthogonalization — in **opposite order**. Empirically Muon wins, yet prior theory says nothing about why. Three pipelines for the update ΔW_t :

Pre-polar (Muon) $\Delta W_t \propto \mathcal{O}(M_t)$

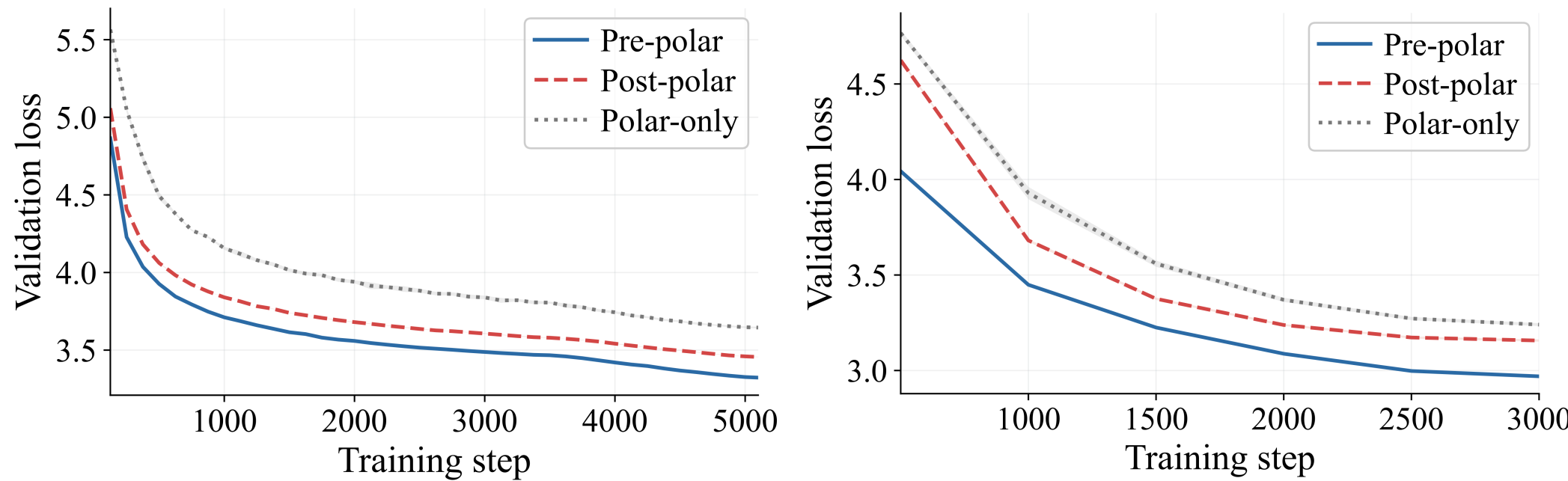
average $\rightarrow$ orthogonalize $\checkmark$

Post-polar (Orth-SGDM) $\tilde{M}_t = \beta \tilde{M}_{t-1} + (1-\beta)\mathcal{O}(G_t)$

orthogonalize $\rightarrow$ average

Polar-only $\Delta W_t \propto \mathcal{O}(G_t)$

no momentum



NanoGPT. Pre-polar wins.

LLaMA 350M. Same trend.

What role does momentum play on the gradient stream, and why does placing it before orthogonalization matter inside Muon?

2 Setup: signal-plus-perturbation gradient model

Momentum buffer. $M_t = \beta M_{t-1} + (1-\beta)G_t$, effective window $T = \frac{1}{1-\beta}$.

Polar factor. $\mathcal{O}(X) = UV^\top$ from the thin SVD $X = U\Sigma V^\top$ — keeps directions, sets all singular values to 1.

Structured gradient (Assumption 1)

$$G_t = G_t^{\text{sig}} + \Xi_t, \quad G_t^{\text{sig}} = \sum_{k=1}^r \lambda_k(t) u_k v_k^\top$$

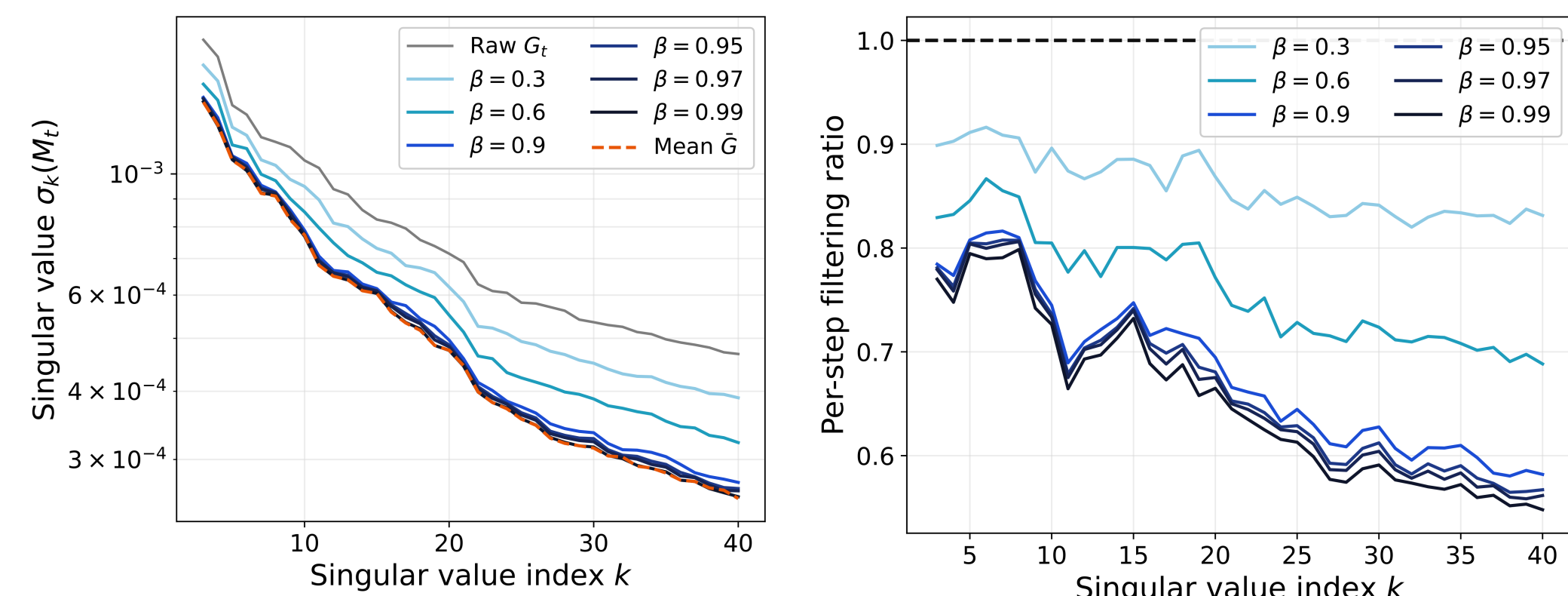
- **Coherent signal** — directions $\{u_k v_k^\top\}$ are time-invariant.
- **BVMZOS perturbation** — $\mathbb{E}[\Xi_t] = 0$, $\mathbb{E}\|\Xi_t\|_F^2 \leq \eta$, uncorrelated across t (covers heavy-tailed, time-dependent noise).
- **Signal persistence** — filtered coordinate $|\bar{\lambda}_k(t)| \geq c_{\text{sig}} \lambda_k$ stays away from zero.

3 Momentum opens a spectral gap

Theorem 1 · w.p. $\geq 1-(2T-1)^{-1/2}$

$$\underbrace{\sigma_k(M_t) \geq c_{\text{sig}} \lambda_k - \frac{\sqrt{\eta}}{(2T-1)^{1/4}}}_{\text{head } k \leq r: \text{ signal kept}}, \quad \underbrace{\sigma_{r+1}(M_t) \leq \frac{\sqrt{\eta}}{(2T-1)^{1/4}}}_{\text{tail: noise} \rightarrow 0}$$

The gap **widens with the window** T (heavier β); the tail noise floor decays as $T^{-1/4}$ — the *opposite* of classical analyses that degrade at large β .



Filtered spectra. As $\beta \uparrow$, the momentum spectrum moves toward the mean-gradient spectrum $\tilde{G}$ — head first, tail later.

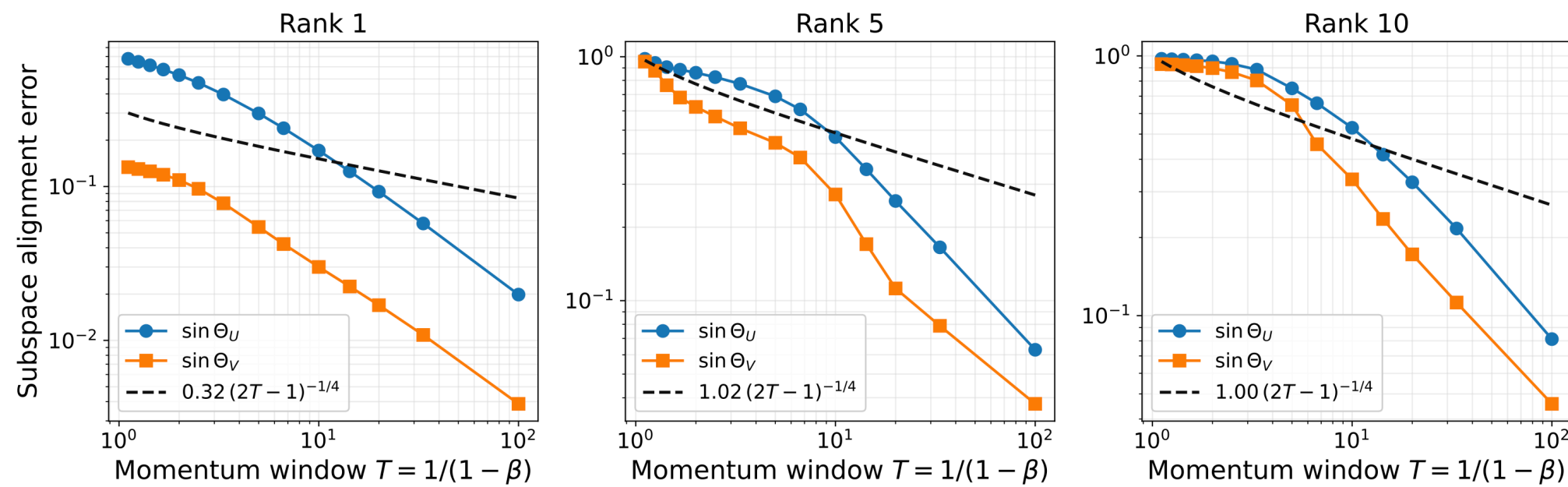
Per-step filtering ratio $\sigma_k(M)/\sigma_k(G)$. The head-vs-tail gap is exactly the spectral gap of Theorem 1: head preserved, tail suppressed.

4 Singular-subspace reliability

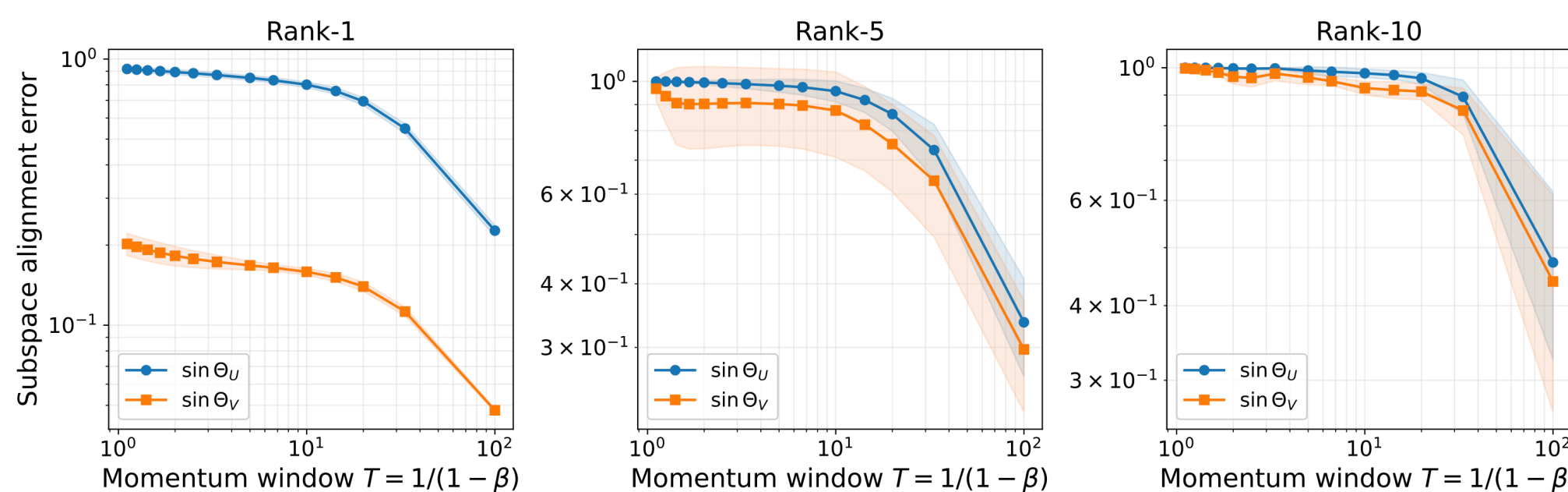
Corollary 1 (via Wedin $\sin \Theta$)

$$\|\sin \Theta(\hat{U}_t, U)\|_2, \|\sin \Theta(\hat{V}_t, V)\|_2 \leq \frac{\sqrt{\eta}/(2T-1)^{1/4}}{c_{\text{sig}} \lambda_r - \sqrt{\eta}/(2T-1)^{1/4}} = O(T^{-1/4})$$

The larger the gap, the closer M_t 's leading subspaces are to the signal's — so the polar factor acts on a **reliable** matrix.



Stationary NanoGPT probe. Subspace errors $\sin \Theta_U, \sin \Theta_V$ at ranks $r \in \{1, 5, 10\}$ fall with T .



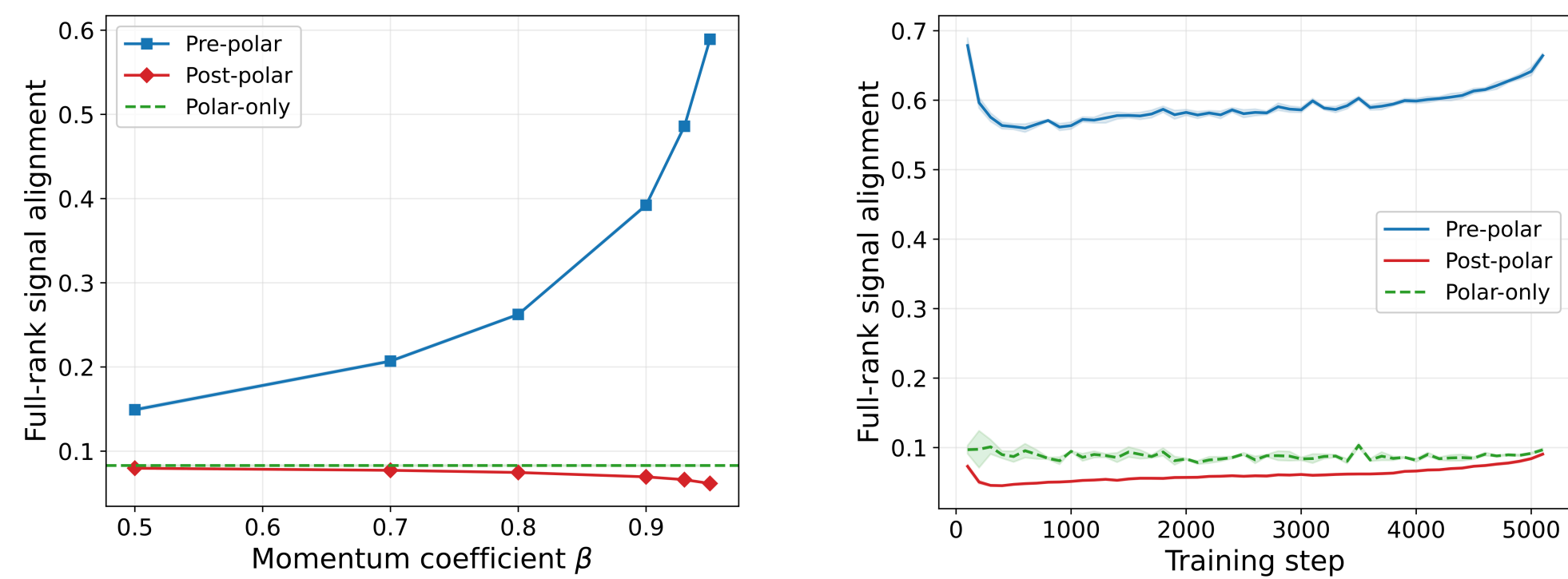
Trajectory probe (live training). Subspace alignment error decays in T at every rank, even as the model trains. Corollary 1's prediction survives the time-varying signal of live training.

5 Pre-polar dominates Post-polar and Polar-only

Theorem 2 · Pre-polar dominates (signal alignment, large T)

$$\underbrace{\mathbb{E} \left[\frac{\langle \mathcal{O}(M_t), G^{\text{sig}} \rangle_F}{\langle \mathcal{O}(G^{\text{sig}}), G^{\text{sig}} \rangle_F} \right]}_{\text{Pre-polar (Muon)}} \geq \max \left\{ \underbrace{\mathbb{E} \left[\frac{\langle \mathcal{O}(G_t), G^{\text{sig}} \rangle_F}{\langle \mathcal{O}(G^{\text{sig}}), G^{\text{sig}} \rangle_F} \right]}_{\text{Polar-only}}, \underbrace{\mathbb{E} \left[\frac{\langle \tilde{M}_t, G^{\text{sig}} \rangle_F}{\langle \mathcal{O}(G^{\text{sig}}), G^{\text{sig}} \rangle_F} \right]}_{\text{Post-polar}} \right\} + \frac{C}{2}$$

The polar factor replaces every singular value with **1**, so acting directly on a noisy gradient **erases the amplitude gap** between signal and noise directions — and subsequent averaging cannot recover what was already destroyed.



Live NanoGPT, vs β . Pre-polar climbs; others nearly flat.

Across training ($\beta = 0.95$). Pre-polar leads throughout.

6 Quantitative low-SNR signal-recovery separation

Theorem 3 · rank-1 spiked Gaussian, low SNR $\lambda < 0.25 \sigma_\Xi (\sqrt{m} - \sqrt{n})$

$$\underbrace{\mathbb{E} \left[\frac{\langle \mathcal{O}(G_t), G^{\text{sig}} \rangle_F}{\langle \mathcal{O}(G^{\text{sig}}), G^{\text{sig}} \rangle_F} \right]}_{\text{Polar-only alignment}} \leq \underbrace{\frac{1}{\sqrt{m}} + \frac{4\lambda}{\sigma_\Xi (\sqrt{m} - \sqrt{n})}}_{\xrightarrow{m \rightarrow \infty} 0} + 4\sqrt{n} e^{-c(\sqrt{m} - \sqrt{n})^2/4}$$

Polar-only alignment vanishes as $m \rightarrow \infty$ — the rank-1 signal is buried in the noise bulk. **Pre-polar still recovers it** for a large enough window T (Thm 2).

Takeaways

- **Momentum is a spectral filter on the gradient stream.**
- **Denoise first, then orthogonalize.** The polar factor benefits from operating on a buffer whose spectrum has already been denoised.
- **The mechanism extends to any matrix-based optimizer beyond Muon.** Spectral filtering is a property of momentum acting on any noisy matrix gradient stream.